MSG No. 51.297 $Pm'm'a'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `2a`, site symmetry: `.2/m'.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,6,8]`

Table 2: Wyckoff site: `2b`, site symmetry: `.2/m'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[2,4,6,8]`

Table 3: Wyckoff site: `2c`, site symmetry: `.2/m'.`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,4,6,8]`

Table 4: Wyckoff site: `2d`, site symmetry: `.2/m'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,3,5,7]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,4,6,8]`

Table 5: Wyckoff site: `2e`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, 0, -z\right]$	`[2,3,5,8]`

Table 6: Wyckoff site: `2f`, site symmetry: `m'm'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	`[1,4,6,7]`
2	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	`[2,3,5,8]`

Table 7: Wyckoff site: `4g`, site symmetry: `.2.`

No.	position	mapping
1	$[0, y, 0]$	`[1,3]`
2	$\left[\frac{1}{2}, -y, 0\right]$	`[2,4]`

continued ...

Table 7

No.	position	mapping
3	$[0,\ -y,\ 0]$	`[5,7]`
4	$\left[\frac{1}{2},\ y,\ 0\right]$	`[6,8]`

Table 8: Wyckoff site: `4h`, site symmetry: `.2.`

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	`[1,3]`
2	$\left[\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	`[2,4]`
3	$\left[0,\ -y,\ \frac{1}{2}\right]$	`[5,7]`
4	$\left[\frac{1}{2},\ y,\ \frac{1}{2}\right]$	`[6,8]`

Table 9: Wyckoff site: `4i`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,7]`
2	$\left[x+\frac{1}{2},\ 0,\ -z\right]$	`[2,8]`
3	$[-x,\ 0,\ -z]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	`[4,6]`

Table 10: Wyckoff site: `4j`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	`[1,7]`
2	$\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[2,8]`
3	$\left[-x,\ \frac{1}{2},\ -z\right]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	`[4,6]`

Table 11: Wyckoff site: `4k`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	`[1,6]`
2	$\left[\frac{3}{4},\ -y,\ -z\right]$	`[2,5]`
3	$\left[\frac{3}{4},\ y,\ -z\right]$	`[3,8]`
4	$\left[\frac{1}{4},\ -y,\ z\right]$	`[4,7]`

Table 12: Wyckoff site: `8l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x+\frac{1}{2},\ -y,\ -z\right]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`

continued ...

Table 12

No.	position	mapping
4	$\left[\frac{1}{2} - x, -y, z\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[\frac{1}{2} - x, y, z\right]$	[6]
7	$[x, -y, z]$	[7]
8	$\left[x + \frac{1}{2}, y, -z\right]$	[8]